

ACR Appropriateness Criteria® Methodology

David A. Kurth, MA, MPH^a, Boaz K. Karmazyn, MD^b, Christine A. Waldrip, RN, MHA^c,
Mythreyi Chatfield, PhD^d, Mark E. Lockhart, MD, MPH^e

Abstract

The ACR Appropriateness Criteria® (AC) are evidence-based guidelines that guide physicians on appropriate image ordering. The AC development and revision process follows a transparent methodology that includes the systematic analysis of current medical literature from peer-reviewed journals and the application of well-established guidelines standards (the Institute of Medicine's *Clinical Practice Guidelines We Can Trust*) and methodologies (the RAND/UCLA Appropriateness Method and Grading of Recommendations Assessment, Development and Evaluation) to rate the benefits and potential risks, or appropriateness, of imaging and treatment procedures for specific clinical scenarios. In the October 2020 release, the methodology is applied in the development of 198 AC documents covering 1,760 clinical scenarios to make more than 8,815 recommendations, authored by more than 600 members representing multiple expert societies, and using more than 6,200 references. The ACR is recognized as a qualified provider-led entity by CMS for the development of appropriate use criteria. This paper describes the methodology and illustrates adherence to the process in the development of the AC.

Key Words: Appropriateness Criteria, appropriate use criteria, AUC, methodology, clinical practice guidelines

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INTRODUCTION

The basis for any evidence-based methodology is transparency, standardization, and continuous improvement. This report captures years of refinement of the methodologic process of the ACR Appropriateness Criteria® (AC). Concepts and methods from standard methodologies and

regulatory requirements are used to structure the AC development methodology [1-21]. The ACR AC address 198 clinical conditions with 965 variants and 1,760 clinical scenarios in the October 2020 release and use current peer-reviewed evidence to identify appropriate diagnostic imaging or therapeutic procedures. New topics are added to reflect changes in technology and clinical practice. AC are reviewed annually to assess whether new and important scientific evidence warrants topic revision or development.

AC content is developed by panels of physicians, grouped by clinical area, with support from ACR staff members who are trained in evidence-based guideline development. The methodology is based on the *RAND/UCLA Appropriateness Method User's Manual* [22]. With the formation of the ACR AC Methodology Subcommittee in 2011, the ACR moved to more formal guidelines development methodologies and processes informed by evidence-based practice centers, the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach [23], and the standards for evidence-based guidelines development from the Institute of Medicine's [12] publication *Clinical Practice Guidelines We Can Trust* (Table 1).

The Guidelines & Recommendations—Authoring, Validation & IT Access System (GRAVITAS) is a web-based proprietary platform used to manage the AC

^aSenior Director of Guidelines Development, American College of Radiology, Reston, Virginia.

^bRiley Hospital for Children, Indiana University, Indianapolis, Indiana.

^cDirector of Appropriateness Criteria, American College of Radiology, Reston, Virginia.

^dExecutive Vice President for Quality and Safety, American College of Radiology, Reston, Virginia.

^eChair, Radiology Departmental Appointments, Promotions, and Tenure Committee, and Departmental Chief, Genitourinary Imaging, University of Alabama at Birmingham, Birmingham, Alabama; ACR Chair, Appropriateness Committee; Chair, Society of Radiologists in Ultrasound, Annual Meeting Program Committee; and Chair, Research Committee of AIUM Future Fund.

Corresponding author and reprints: David A. Kurth, MA, MPH, American College of Radiology, 1891 Preston White Drive, Reston, VA; e-mail: dkurth@acr.org.

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Table 1. Compliance with Institute of Medicine standards

Standard 1: Establishing transparency	Posted on the public area of the ACR website at http://www.acr.org/Quality-Safety/Appropriateness-Criteria
Standard 2: Management of COI	All potential conflicts are disclosed and updated annually. ACR legal and compliance staff members review and remediate conflicts according to written ACR COI policy.
Standard 3: Guideline development group composition	AC panels include multidisciplinary experts and clinicians representing more than 40 medical specialty societies.
Standard 4: Clinical practice guideline–systematic review intersection	Structured literature search based on key words and limits conducted by ACR staff members Articles are dispositioned and search is modified as needed. Staff members evaluate the quality of selected evidence using standard criteria, and SOE is calculated using a standard algorithm.
Standard 5: Establishing evidence foundations for and rating strength of recommendations	Panel members rate the benefits and harms (appropriateness) on the basis of the narrative summary of evidence. Individual ratings are consolidated into an appropriateness rating using a standard method. Evidence associated with recommendations is documented in the evidence table.
Standard 6: Articulation of recommendations	Recommendations are stated for each imaging procedure and clinical scenario as usually appropriate, may be appropriate, or usually not appropriate. Where the procedure may be appropriate, comments clearly state under what circumstances the procedure would be appropriate.
Standard 7: External review	External review is sought at the end of each publication cycle, and feedback is incorporated into the next annual review.
Standard 8: Updating	All ACR ACs are reviewed annually and updated as needed on the basis of a review of the current literature.

Note: AC = Appropriateness Criteria; COI = conflict of interest; SOE = strength of evidence.

workflows and digital content, such as the systematic literature search process.

DOCUMENT DEVELOPMENT AND REVISION

Prioritizing Content Development: Annual Review

The chairs of the expert panels determine the topics that require revision and development of new topics during an annual review process. Revision of an existing topic is considered if new evidence is available that can change the recommendation for an imaging procedure or can identify an important content gap. New topic proposals can be suggested by the members of the panel, customers of the clinical decision support mechanism, and any interested person who submits a proposal on the ACR website. New content development and revisions to existing content follow the AC methodology.

Selecting Topics

Topics constitute the clinical scope of the document. New topic selection has evolved over the years from a focus on initial presentation of the clinical scenario to the current expanded approach to address imaging at a variety of points along the scenario's care pathway, including initial workup, diagnosis (including staging), treatment, and follow-up. Also, the topic selection process has become more inclusive and systematic in recent years, incorporating feedback from a variety of sources.

Defining Variants

Variants represent clinical and demographic factors that can affect the appropriate clinical decision and may include, as applicable, the method of imaging (eg, initial, follow-up, screening, surveillance), patient demographic and clinical characteristics, care setting (eg, outpatient, inpatient), and prior diagnostic (imaging or laboratory) test results. A topic

typically has several variants that are expected to cover the most common groups of patients' characteristics that may affect the selection of imaging procedure.

A topic may contain variant descriptions that do not require imaging as well as those that may. For example, in the topic of low back pain, the variant description "uncomplicated low back pain" would likely recommend that no imaging be performed. Another variant description may be "low back pain with prior lumbar surgery," for which imaging may be appropriate.

Variant definition is refined with input from the panel, including relevant expert clinician representatives, and is approved by the panel chairs before the systemic literature search. Variants may be further refined as panel members review the draft of the narrative.

Gathering Evidence

For each AC topic, the author evaluates available current peer-reviewed medical literature related to the topic and its variants before updating or developing the AC document. ACR staff members document search strategies for transparency and replicability.

Systematic Literature Search Process

ACR staff members search the English-language, peer-reviewed literature available in PubMed using key terms and search parameters identified by the topic author on the basis of the variants. Case reports are excluded from the search. Panel members are encouraged to include clinical practice guidelines and consensus statements that are available and to present relevant findings or contradictions between recommendations. However, guidelines are considered secondary evidence sources, providing low-quality evidence only. Authors evaluate results from the systematic search to select the most relevant articles to cite for the topic and variants. Studies identified by the search strategy but not included are documented and dispositioned in GRAVITAS. Authors have minimal flexibility to add articles that may not be identified in a systematic search, such as those that are new, classic or seminal articles, or articles with incorrect Medical Subject Heading terms or missing keywords.

Assessing Evidence

The topic author assesses the literature, then drafts or revises the narrative summarizing the evidence. ACR staff members rate the study quality for each article included in the narrative and drafts an evidence table (ET).

The GRADE approach is the basis of the concepts used to evaluate the evidence. The GRADE working group's

mission is to resolve the confusion among the different systems of rating evidence and recommendations [23,24].

The strength of evidence (SOE) for a recommendation is based on the number of quality studies and the consistency of the study results associated with the recommendation, as well as the directness of the evidence to the recommendation. This is based on the GRADE approach to evaluate evidence that mentions factors that decrease the quality of the evidence [23], such as poor-quality randomized controlled trials (RCTs), inconsistency of results, indirectness of evidence, and sparse evidence. The ACR created explicit criteria consistent with this methodology to improve consistency among those evaluating the quality of the references, to provide transparency, and to augment the clarity of the assessment by the panels.

Study Quality

Grading the evidence starts with the assessment of the references' study quality. On the basis of the GRADE approach [23,24], studies are assessed for components that may affect bias. RCTs, when designed and implemented well, provide high-quality evidence. Observational studies may also provide high-quality evidence for diagnostic studies, but GRADE recommends the assessment of different study quality components for observational studies and RCTs [24]. Similarly, the ACR uses different study quality components for diagnostic, therapeutic, and meta-analytic studies. Staff members evaluate each peer-reviewed journal article cited for the appropriate components. The use of a standardized algorithm to determine the study quality category helps ACR staff members ensure reliable, repeatable, and systematic assessments of the study quality.

Determining Study Quality Category

Study quality components correspond to the study type (diagnostic, therapeutic, or meta-analytic). The assumption is that components not mentioned or not implemented in the study potentially may introduce bias. The study quality category is assigned by determining the study type and assessing the corresponding eight equally weighted quality components in the study [25] (Table 2) and is documented in the ET for every citation used to support a recommendation. The quality of diagnostic and therapeutic studies is categorized from 1 (controls for common biases) to 4 (cannot be used as a primary evidence; see Table 3). Meta-analytic studies are categorized as good (effective analysis that leads to valid results and conclusions) or inadequate (lacking key components that may bias the results and conclusions; see Table 4). Study quality assessments on more than 6,200 references support the current AC.

Table 2. Study quality components by study type

Diagnostic Studies	Therapeutic Studies	Meta-Analytic Studies
Uncertainty measure included in statistical analysis of results	Statistical measure reported	At least two studies included in the statistical analysis
Prospective: The study was designed prior to data collection	Blinding as relevant to reported outcomes and ethically allowable	Clear purpose stated in combining studies
Systematic recruitment or consecutive series to reduce selection bias	Uncertainty measure included in statistical analysis of results	Prospective: The study was designed prior to the hypothesis formation, study selection, and analytic strategies
Standard of reference , preferably gold standard, clearly defined	Prospective: The study was designed prior to the data collection	Inclusion/exclusion criteria determined prospectively
Reference standard applied consistently to all subjects	Allocation of subjects into distinct test and control groups	Inclusion criteria appropriate for questions asked
At least two independent readers to reduce bias introduced by the ability of the readers of the diagnostic tests	Random allocation of subjects to ensure representative samples; each subject has the same chance of receiving any of the possible interventions.	Pooled statistic is calculated to obtain a more precise estimate than any single study
Index test results blinded from reference standard	Length of follow-up explicitly stated	Study designs of included studies appropriate for analysis question
Reference standard results blinded from index tests	Disposition of all subjects enrolled in study	Study results of included studies must be comparable for aggregation

Assessing SOE

The SOE is an assessment of the evidence quality for each recommendation or the extent to which one can be confident that an estimate of effect is correct [4].

An algorithm is used to categorize the SOE as strong, moderate, limited, expert consensus, or expert opinion. The algorithm assesses the number of high-quality studies whose results are consistent among the studies assigned to the recommendation and directly related to the clinical condition and procedure of the recommendation. At least two high-quality studies (category 1, 2, or “good”) that have consistent results that are related directly to the recommendation meet the definition of strong SOE. If there is (1) indirectness of the results to the recommendation, (2) inconsistent results among the high-quality studies, or (3) only one high-quality study, the SOE is categorized as moderate. If there are no high-quality studies or the high-quality studies are indirect and inconsistent, the SOE is characterized as limited. When there is no citation associated with a recommendation for which the panel agrees on the appropriateness category, the SOE is expert consensus. When the panel cannot agree on the recommendation, the SOE is expert opinion, regardless of number, study quality,

indirectness, or inconsistency of the studies associated with the recommendation. The assigned SOE is reviewed and, if necessary, may be revised by authors and panel members.

Indirectness is considered present when the results of all the studies assigned to a recommendation are not directly related to the clinical condition or procedure of that recommendation. To have directness, all high-quality references (category “good,” 1, or 2) associated with the recommendation must have results directly related to the clinical condition and procedure for that recommendation.

Inconsistency is considered present when the results of all the studies associated with the recommendation have a different magnitude or direction. The magnitude of the results is dependent on the measure used, but it should be obvious to most readers that there is a difference in the study results. The cited studies that are assessed for consistency should have comparable study design rigor, hypotheses, data collection, and analytic methods so that their results can be compared. Consistency is determined when all the relevant cited high-quality studies demonstrate the same effect; that is, they have similar results in terms of magnitude and direction.

The author’s summary in the narrative interprets the results for the studies associated with the recommendation and

Table 3. Definitions of diagnostic and therapeutic study quality categories

Study Quality Category	Study Quality Category Definition	Criteria for Diagnostic Studies	Criteria for Therapeutic Studies
Category 1	The study is well designed and accounts for common biases.	The source has all 8 diagnostic study quality elements.	The source has at least 5 therapeutic study quality elements.
Category 2	The study is moderately well designed and accounts for most common biases.	The source has 6 or 7 diagnostic study quality elements.	The source has 3 or 4 therapeutic study quality elements.
Category 3	The study has important study design limitations.	The source has 3, 4, or 5 diagnostic study quality elements.	The source has 1 or 2 therapeutic study quality elements.
Category 4	The study or source is not useful as primary evidence. The article may not be a clinical study, the study design is invalid, or conclusions are based on expert consensus. The study does not meet the criteria for or is not a hypothesis-based clinical study (eg, book chapter, case report, case series description); or The study may synthesize and draw conclusions about several studies, such as a literature review article or book chapter, but is not primary evidence; or The study is an expert opinion or consensus document.	The source has 0, 1, or 2 diagnostic study quality elements.	The source has 0 therapeutic study quality elements.

the directness and consistency of the results. [Figure 1](#) illustrates the SOE category distribution by appropriateness category and panel for more than 60% of the recommendations in the October 2020 release. Thirty-nine percent of the recommendations do not have calculated SOE because the topics are new, are under review, or were calculated using the original SOE method. As the topics are revised, the SOE for these recommendations will be assessed.

[Figure 1](#) also suggests that strong evidence is cited most often for procedures that are usually appropriate. This may reflect some publication bias in that most studies provide evidence supporting the performance of an effective procedure rather than disproving a procedure's value [26]. Less rigorous evidence in the other categories may also be because of more limited experience in procedures that are usually not appropriate compared with procedures that are usually appropriate.

Documenting the Evidence

The ET summarizes the citations and their quality. ACR staff members draft the ET to maintain consistent, high-quality summaries. After an ET is constructed, it is reviewed by the panel for accuracy, completeness, and validity. The October 2020 release included study quality ratings for 6,241 articles ([Fig. 2](#)). The study quality rating of “quality not assessed” are meta-analysis studies that were cited in pre-2018 topics, before the meta-analysis study quality methodology was resolved. As the topics are revised, the quality of these studies will be assessed.

Drafting Evidence Summaries (AC Topic Narrative)

The AC documents have evolved from descriptive review articles to evidence summary documents. They include analysis of the peer-reviewed study results found through a

Table 4. Definitions of study quality categories for meta-analytic studies

Study Quality Category	Study Quality Category Definition	Criteria for Meta-Analytic Studies
Good	The meta-analytic study is well designed, uses sound methodology, and performs effective analysis that leads to valid results and conclusions. Typically, the analysis provides additional insights or more statistical confidence in the results.	The source has all 8 meta-analytic study quality components.
Inadequate	The study's design, methods, or analysis lacks key meta-analytic study quality components that may bias the results and conclusions.	The source has fewer than 8 meta-analytic study quality components.

systematic literature search and align evidence to corresponding imaging recommendations for specific clinical scenarios.

Evaluating Benefits and Potential Risks (Appropriateness Rating)

The ACR defines appropriateness as a measure of whether expected health benefit exceeds expected negative consequences sufficiently to justify imaging, without

consideration of cost, a definition adopted from the *RAND/UCLA Appropriateness Method User's Manual* [22].

The appropriateness rating process uses a modified Delphi method to determine the panel's recommendations regarding interpretation of available evidence. Panel members use the evidence summary and rate recommendations on the basis of the balance of benefits and potential risks of performing a radiologic examination or performing an imaging-guided procedure for a clinical scenario. Ratings are

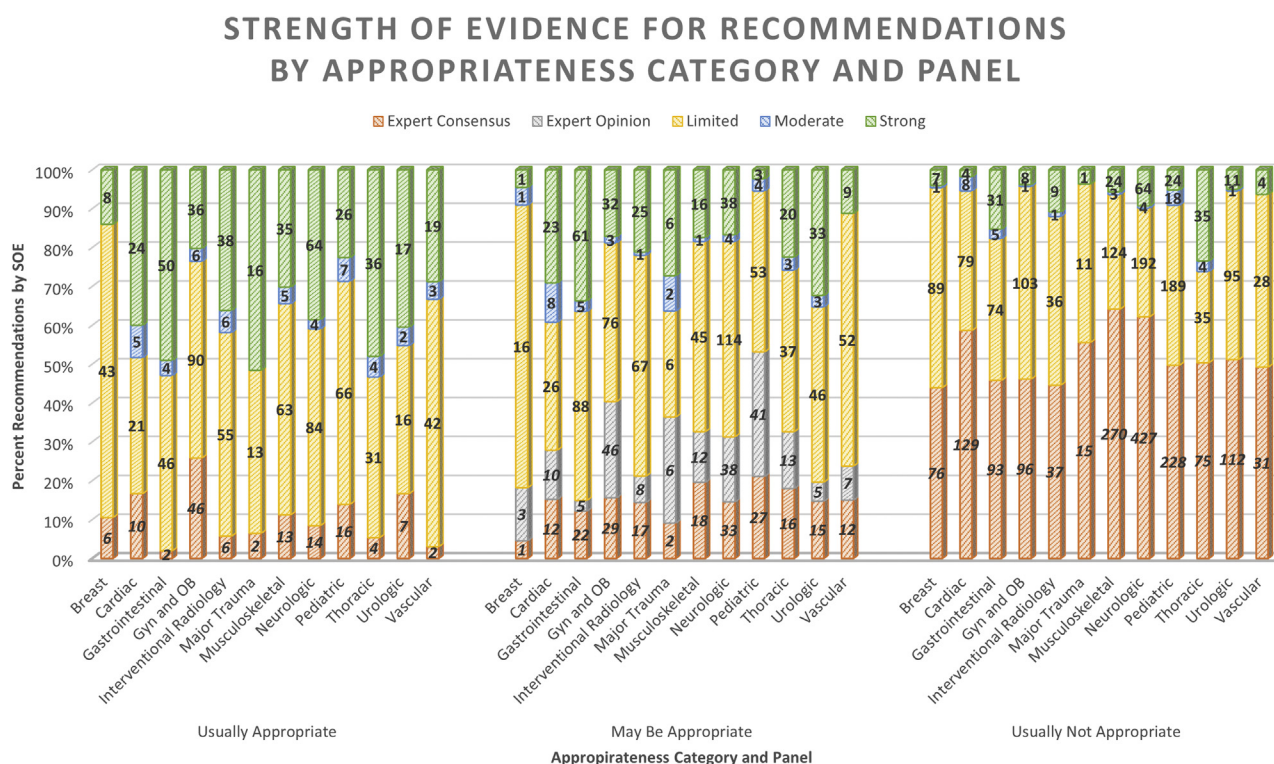


Fig. 1. Strength of evidence assessment. This chart illustrates the distribution of strength of evidence for the 8,815 recommendations in the October 2020 release. Gyn = gynecology; OB = obstetrics.

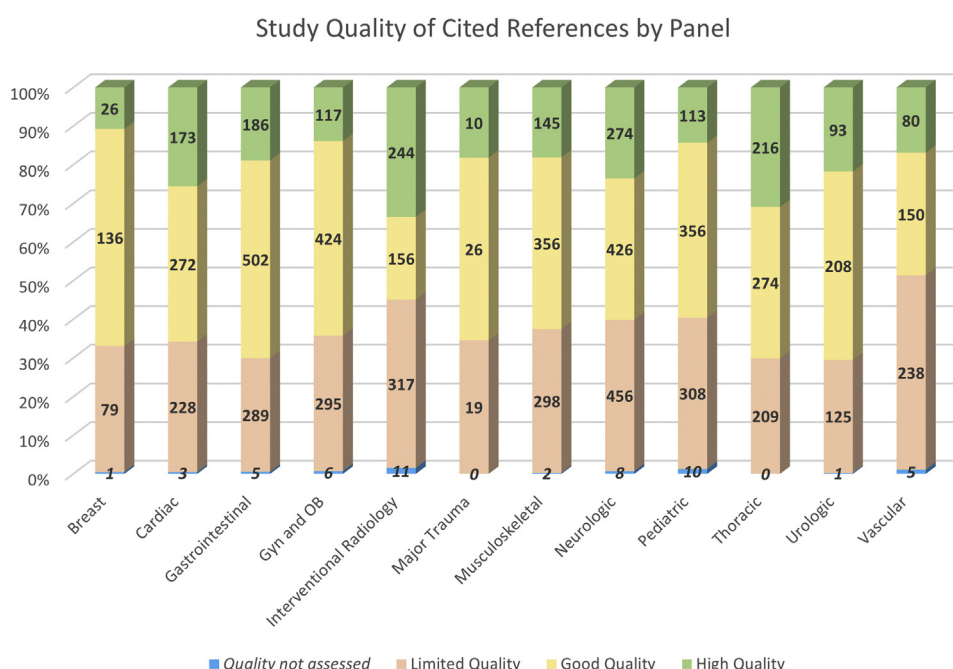


Fig. 2. Study quality categories. This chart illustrates the study quality of the 6,241 references that inform recommendations. Some references are used by more than one panel. Gyn = gynecology; OB = obstetrics.

anonymized to allow experts with different clinical expertise to indicate their interpretations of the evidence. The consensus methodology does not diminish the evidence for the recommendations; it is merely a method to objectively determine group consensus. Furthermore, the AC methodology displays the strength of the evidence assigned to each recommendation.

The AC are population-based recommendations and are not intended to be absolute recommendations applicable to

every possible clinical presentation, because there are numerous patient-specific health and preference characteristics and health provider expertise, experience, and availability that affect care.

Assumptions

Assumptions are made with regard to what may be included in the consideration of appropriateness that facilitate reliable ratings among the panel members (Table 5).

Table 5. List of assumptions relevant to appropriateness rating

Assumptions	Discussion
1. Relative radiation levels	When rating appropriateness, the panelist does not consider the relative radiation level of the procedure or treatment. Relative radiation levels may be considered when assessing risks and benefits only when other medical imaging examinations or procedures have nearly equivalent diagnostic accuracy or test performance.
2. Expertise	When rating appropriateness, the panelist assumes that all procedures or treatments are performed by an expert.
3. Availability	When rating appropriateness, the panelist assumes that all procedures or treatments listed in the variant tables are available.
4. Procedures that are contraindicated	When rating appropriateness, the panelist assumes that the patients, as defined by the topic and variants, do not have contraindications to any of the procedures or treatments.
5. Cost of the procedure	When rating appropriateness, the panelist does not consider the cost of the procedure or treatment.

Rating Appropriate Use

The appropriateness rating methodology systematically and transparently captures panels' recommendations while mitigating undue influence of one or more panel members on others. The panels rate appropriateness for a recommendation by using the evidence presented in the narrative. The median of the individual panel members' ratings is the panel's rating for the recommendation. The assessment of the deviation of individual ratings from the panel rating determines whether there is disagreement among the panel regarding the rating. When the interpercentile range, which is a statistical measure of dispersion of a distribution, is greater than the interpercentile range adjusted for symmetry, there is disagreement [22].

In assessing the harms and benefits, panel members focus on how well the imaging procedure can provide clinical information to move the patient along the care pathway to the best outcomes. This should be based primarily on the available evidence and its quality related to similar patient population as defined by the variants. Panel members will often need to apply their expertise to evaluate the accuracy and benefit of the imaging procedure for the variant on the basis of evidence for populations and clinical scenarios that may have different characteristics.

The appropriateness rating range is an ordinal scale that uses integers from 1 to 9 grouped into three categories. At the ends of the scale are the "usually not appropriate" category, which describes when the potential risks of doing a procedure outweigh the benefits, and the "usually appropriate" category, which describes when the benefits of doing a procedure outweigh the potential risks. The middle "may be appropriate" category is applicable when the risks and

benefits are equivocal or unclear, the dispersion of the individual ratings from the group median rating is too large (ie, disagreement), the evidence is contradictory or unclear, or there are special circumstances or subpopulations that are embedded in the variant that confound the benefit-risks assessment. This assessment process was applied to 8,815 recommendations in the October 2020 release (Fig. 3).

Finalizing the Document

The AC Committee, the oversight body for all of the work of the panels, reviews the final AC document for any clinical errors and to confirm that the narrative clearly presents and supports the recommendations. No substantive changes are directly made by the AC Committee. Feedback and recommendations are sent to the entire panel for review, consideration, and, when appropriate, revision.

GOVERNANCE AND ADMINISTRATION

The ACR Committee on Appropriateness Criteria directs the AC development and revision processes. The committee consists of the specialty chairs from each expert panel, the chair of the Subcommittee on Radiation Exposure, the chair of the AC Methodology Subcommittee, the chair of the AC Patient Engagement Subcommittee, a patient representative, and other experts as needed (Fig. 4).

The expert panel is the primary group developing content. Each expert panel consists of ACR member volunteer physicians. Radiologists with additional expertise in nuclear medicine, emergency radiology, and radiation oncology are recruited to participate in selected topics. Clinical experts nominated by medical specialty societies participate in topics

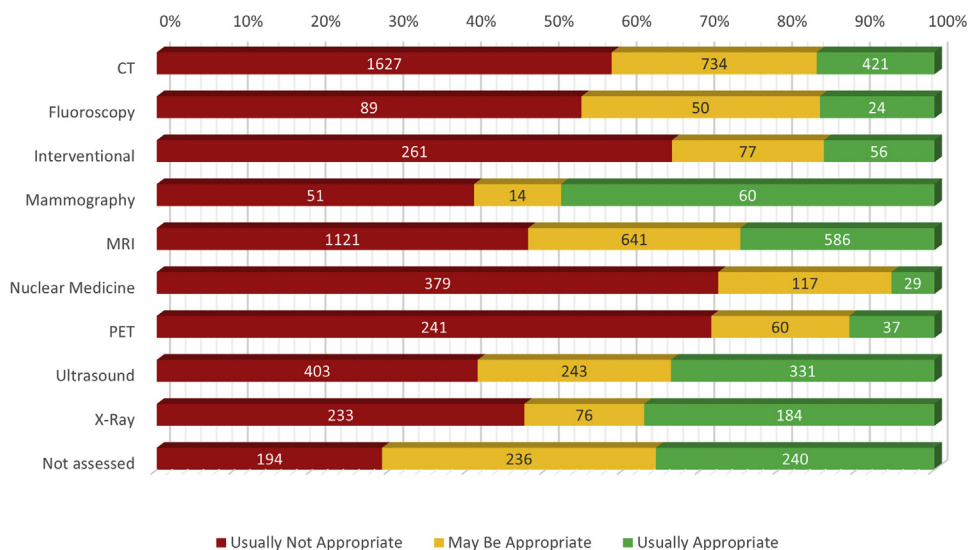


Fig. 3. ACR rating. Distribution of appropriateness rating across the 8,815 recommendations in the October 2020 release.

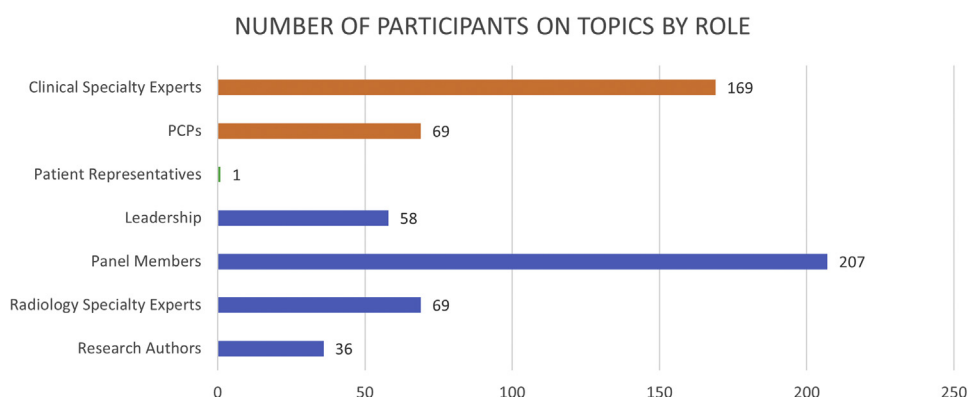


Fig. 4. Composition of panels. Bars in orange represent nonradiologists, bar in green represents nonphysician patient representatives, and bars in blue are radiologists. PCP = primary care physician.

relevant to their practice; participation does not constitute endorsement by their nominating society. AC topics and the oversight committee have representation from primary care, often nominated from national medical societies. The ACR does not seek the endorsement of other medical specialty societies or approval from any entity outside the evidence-based process described here.

Following regulatory requirements, we survey panel members on their skills and expertise to ensure that panels include expertise in the clinical topic, primary care medicine, medical imaging, statistics, and clinical trial design. Panel members' expertise is determined using self-attestation and calculated by the amount of education, training, and experience the member reports for that skill area (Fig. 5).

The AC Methodology Subcommittee assesses and improves the methodology on a continuous basis. It is also

responsible for training materials, process improvement, and education of panel members to uniform standardized implementation of the AC methodology. Members of this subcommittee come from expert panels and have relevant expertise or experience in areas, such as methodology, statistics, and clinical trial design.

Conflict of Interest

Completion and ACR review of the conflict-of-interest (COI) questionnaire and disclosure of potential conflicts are required before participation on the panel. Before each conference call discussing the document, staff members remind participants about the COI policy and to provide updates. Staff members remind panel members to recuse themselves from any discussions on the call when they have potential COIs that have not been reviewed.

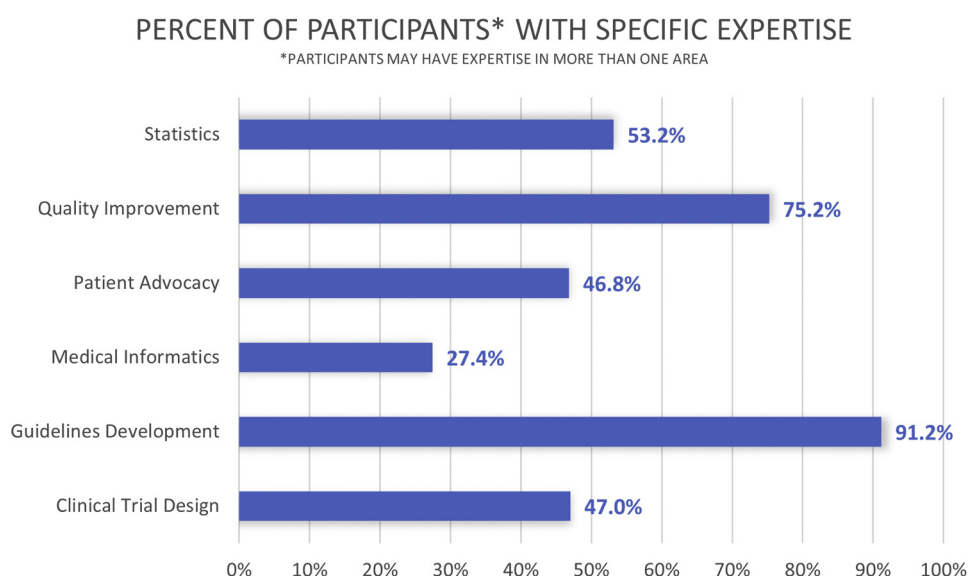


Fig. 5. Expertise covered on panels.

Disclosures are reviewed by ACR compliance and legal staff members according to the current ACR COI policy. The chair of the expert panel is informed of the conflict so that participants with reported COIs may recuse themselves from deliberations related to the COI. Given the highly technological nature of diagnostic imaging, by necessity, we rely on input and analysis from clinical specialists and, in some circumstances, may not be able to exclude everyone with conflicts [12]. The ACR can provide information about reported potential COIs in the AC expert panels on request.

Training and Education

Committee and panel members are required to participate in training and education presentations to ensure adherence to AC methodology and processes.

Methodology summaries are available on the ACR website; these are being developed into webinars and courses with continuing medical education. Summaries include systematic literature searches, evidence assessment (diagnostic study quality, meta-analytic research study quality), rating rounds and assumptions, and SOE.

PUBLICATION AND DISSEMINATION

JACR publishes semiannually a supplement containing new and revised AC documents. *JACR* also publishes selected AC patient summaries in which patients (laypersons) collaborate with the AC Patient Engagement Subcommittee to summarize AC recommendations for patients. These summaries are intended to help patients understand what tests are appropriate for their situations and are designed to help ordering physicians and radiologists better communicate the reasons they are requesting, or performing, particular medical tests.

EXTERNAL FEEDBACK

The ACR releases new and revised AC at least twice a year. The two-month period immediately after each AC release is the formal review and commenting period for external stakeholders (ie, those who are not involved with the content development process). This allows the AC panels to fully review and consider this feedback during the annual review process, which formally incorporates the feedback into the evidence development rather than as an afterthought.

The ACR AC review site (www.review.acr.org) collects feedback and comments systematically and allows staff members to track and respond to commenters. Commenters can address specific issues in existing content, suggest new content for development, and provide feedback related to the methodology.

In addition to the formal review and commenting period, anyone may provide comments on any AC document throughout the year.

SUMMARY AND CONCLUSION

The ACR AC use a transparent, evidence-based methodology that has been implemented to scale in the development of the 198 documents currently available. The methodology continually learns and improves using feedback and new information from all of our stakeholders and communities of guideline developers. Scaling the process without losing standardization depends on automation of the process to the extent possible but is not possible without commitment from all levels of content developers to adhere to the principles of evidence-based guideline development. We hope our experience provides a roadmap that other guideline development processes can follow.

TAKE-HOME POINTS

- The ACR ACs illustrate implementation at scale of an evidence-based guideline development process that can be replicated in other programs.
- ACR ACs are systematic reviews with recommendations that are driven and supported by evidence.
- A modified Delphi process ensures structured compilation of evidence-based assessments of recommendations by clinical experts. It facilitates objective determination of when there is disagreement and provides a voice for all participating experts without undue influence from dominant thought leaders.
- Each ACR AC panel includes representatives from multiple specialties with expertise not just in radiology but also in the relevant clinical area, primary care, clinical trial design, statistics, and methodology.

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